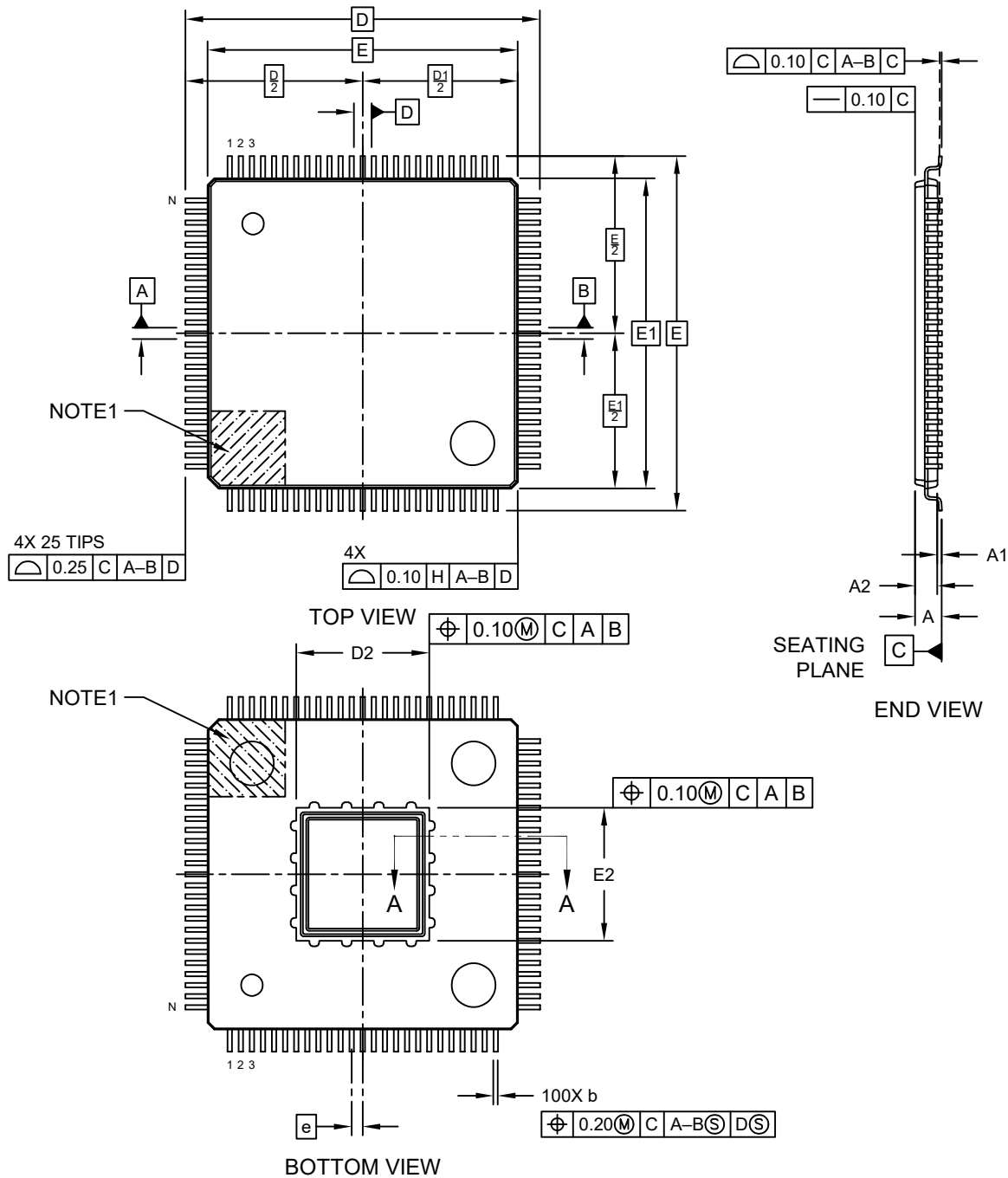


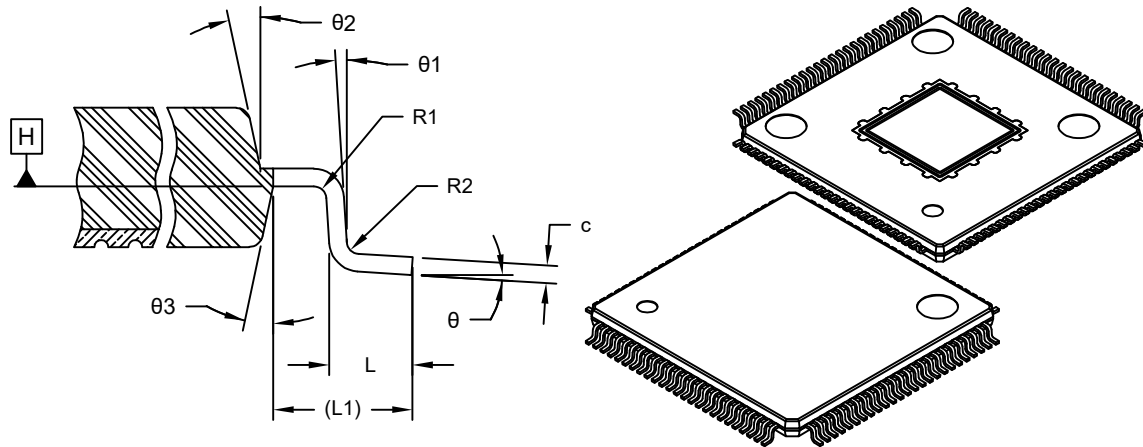
**100-Lead Plastic Thin Quad Flat Pack (4MB) - 14x14x1.0 mm Body [TQFP]  
With 6.0x6.0 mm Grooved Exposed Pad**

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



# **100-Lead Plastic Thin Quad Flat Pack (4MB) - 14x14x1.0 mm Body [TQFP] With 6.0x6.0 mm Grooved Exposed Pad**

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



**SECTION A-A**

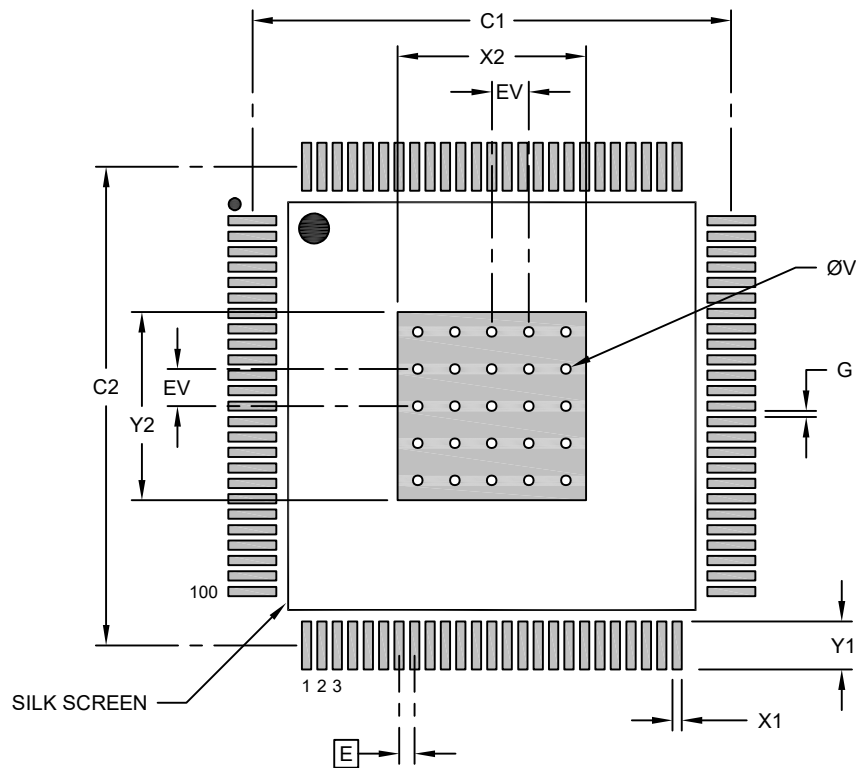
| Dimension Limits         | Units | MILLIMETERS |      |      |
|--------------------------|-------|-------------|------|------|
|                          |       | MIN         | NOM  | MAX  |
| Number of Terminals      | N     | 100         |      |      |
| Pitch                    | e     | 0.50 BSC    |      |      |
| Overall Height           | A     | -           | -    | 1.20 |
| Standoff                 | A1    | 0.05        | -    | 0.15 |
| Molded Package Thickness | A2    | 0.95        | 1.00 | 1.05 |
| Overall Length           | D     | 16.00 BSC   |      |      |
| Molded Package Length    | D1    | 14.00 BSC   |      |      |
| Overall Width            | E     | 16.00 BSC   |      |      |
| Molded Package Width     | E1    | 14.00 BSC   |      |      |
| Exposed Pad Width        | D2    | 5.90        | 6.00 | 6.10 |
| Exposed Pad Width        | E2    | 5.90        | 6.00 | 6.10 |
| Terminal Width           | b     | 0.17        | 0.22 | 0.27 |
| Terminal Thickness       | c     | 0.09        | -    | 0.20 |
| Terminal Length          | L     | 0.45        | 0.60 | 0.75 |
| Footprint                | L1    | 1.00 REF    |      |      |
| Lead Bend Radius         | R1    | 0.08        | -    | -    |
| Lead Bend Radius         | R2    | 0.08        | -    | 0.20 |
| Foot Angle               | θ     | 0°          | 3.5° | 7°   |
| Lead Angle               | θ1    | 0°          | -    | -    |
| Mold Draft Angle         | θ2    | 11°         | 12°  | 13°  |
| Mold Draft Angle         | θ3    | 11°         | 12°  | 13°  |

**Notes:**

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Dimensioning and tolerancing per ASME Y14.5M  
BSC: Basic Dimension. Theoretically exact value shown without tolerances.  
REF: Reference Dimension, usually without tolerance, for information purposes only.

# **100-Lead Plastic Thin Quad Flat Pack (4MB) - 14x14x1.0 mm Body [TQFP] With 6.0x6.0 mm Grooved Exposed Pad**

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



## **RECOMMENDED LAND PATTERN**

| Units                            |    | MILLIMETERS |       |      |
|----------------------------------|----|-------------|-------|------|
| Dimension Limits                 |    | MIN         | NOM   | MAX  |
| Contact Pitch                    | E  | 0.50 BSC    |       |      |
| Center Pad Width                 | X2 |             |       | 6.10 |
| Center Pad Length                | Y2 |             |       | 6.10 |
| Contact Pad Spacing              | C1 |             | 15.50 |      |
| Contact Pad Spacing              | C2 |             | 15.50 |      |
| Contact Pad Width (X100)         | X1 |             |       | 0.30 |
| Contact Pad Length (X100)        | Y1 |             |       | 1.55 |
| Contact Pad to Contact Pad (X96) | G  | 0.20        |       |      |
| Thermal Via Diameter             | V  |             | 0.33  |      |
| Thermal Via Pitch                | EV |             | 1.20  |      |

### **Notes:**

- Dimensioning and tolerancing per ASME Y14.5M  
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
- For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process

Microchip Technology Drawing C04-23548 Rev A